



Predicting Steam Game Prices Using a Manual MLP and Bayesian Modeling

Purpose

Can a game's content, quality, and engagement metrics predict its price on Steam? Pricing is critical for indie developers; digital games have near-zero marginal cost, so price signals perceived value rather than production cost

50K+

Games on Steam

97k+

Observations

41

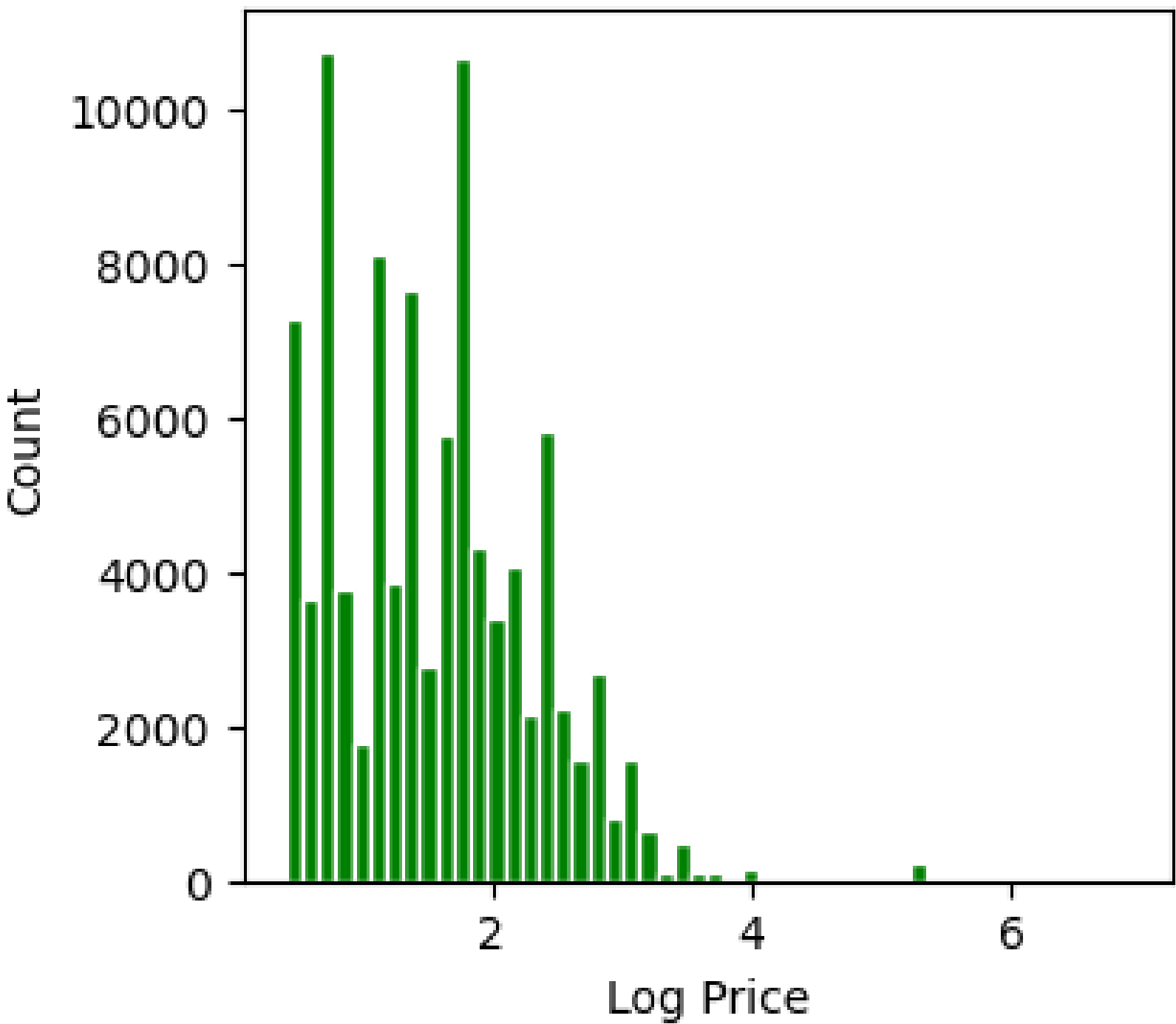
Features

Dataset & Preprocessing

FronkonGames Steam Games Dataset (Hugging Face, Steam API). 120K rows → 97,734 paid games after filtering.

- Removed non-predictive columns; one-hot encoded genres
- Log-transformed skewed features and target (price)
- Engineered: review_ratio, engagement, platform_count
- Min-max scaled on training data only

Distribution of Steam Game Prices (log-scale)



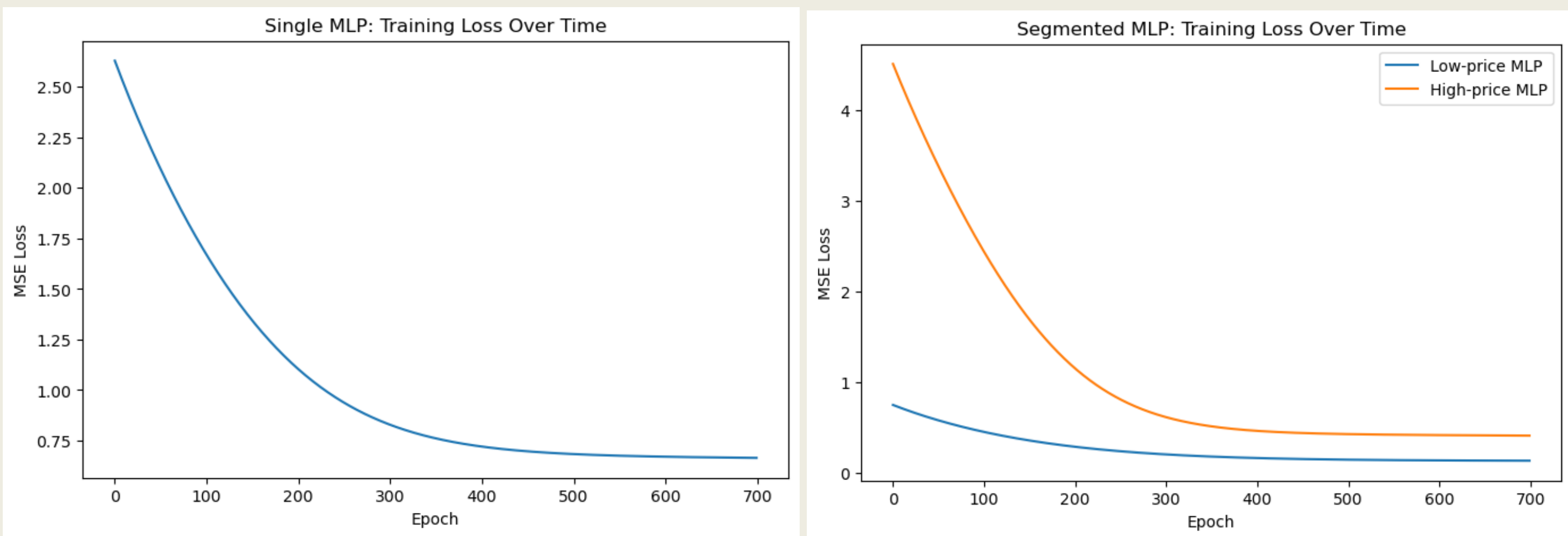
References

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Li et al. (2021). Video game sales via ML. *ISKE 2021*.
Ma (2025). Success drivers across Steam, Twitch, Metacritic. *IJCGT*.
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Model 1: Manual MLP (Python)

Built from scratch in NumPy: 1 hidden layer, 64 units, ReLU, linear output. Full-batch GD, lr = 0.001, MSE loss, gradient clipping ± 5.0 , 700 epochs, 80/20 split.

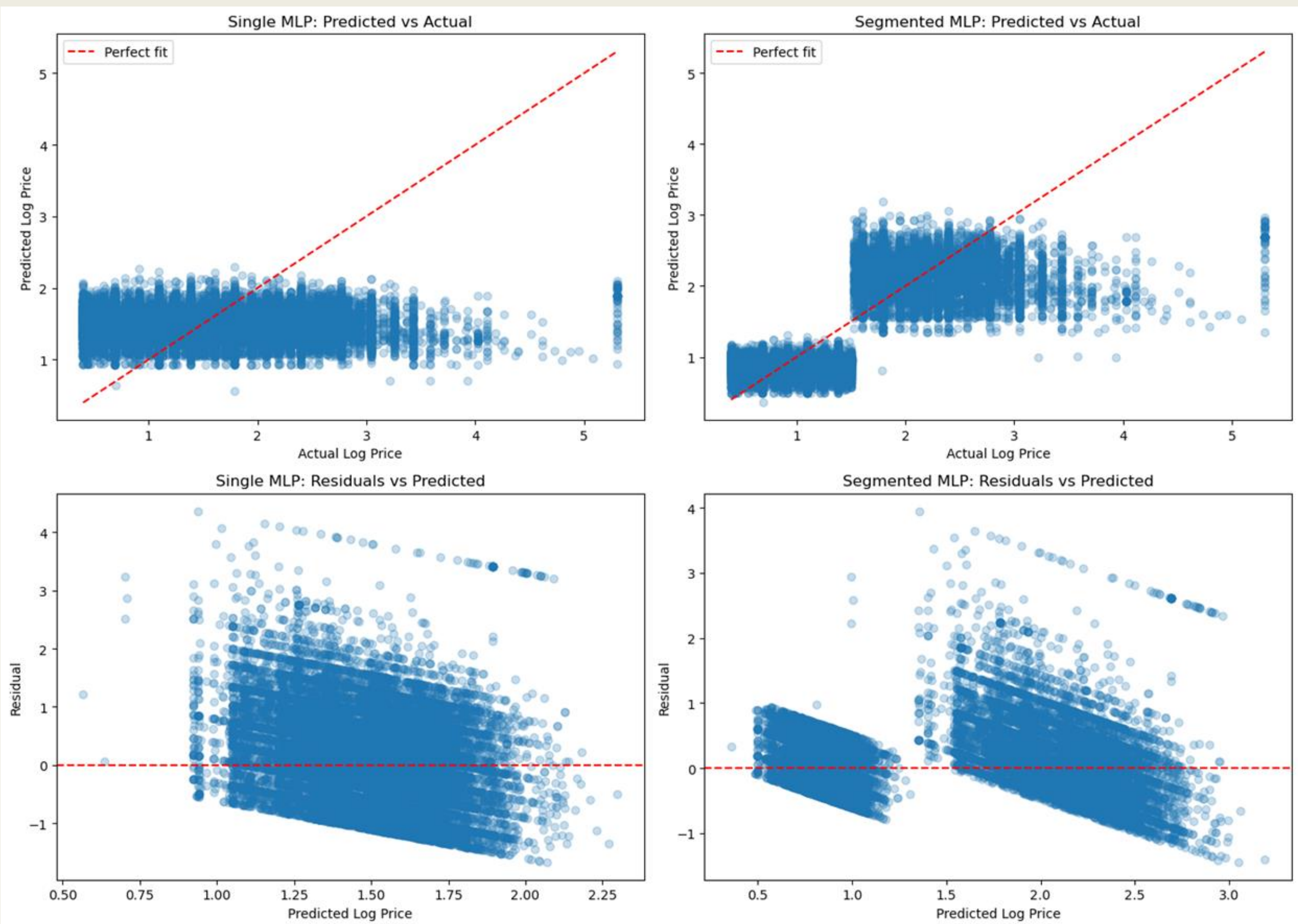
Segmented MLP: Residuals showed large errors for high-priced games. Split training at median log price (1.52); trained two separate MLPs. Test games routed to their segment.



Results (Model 1)

Model	MSE	RMSE	R ²
Single MLP	0.675	0.8218	-0.0849
Low-Price MLP	0.1337	0.3657	-0.1752
High-Price MLP	0.4081	0.6389	-0.3175
Segmented MLP	0.2714	0.5209	0.5640

Single MLP predictions collapsed to a narrow band; segmenting by price regime raised R² from -0.085 to 0.564.



Model 2: Bayesian Log Reg (R)

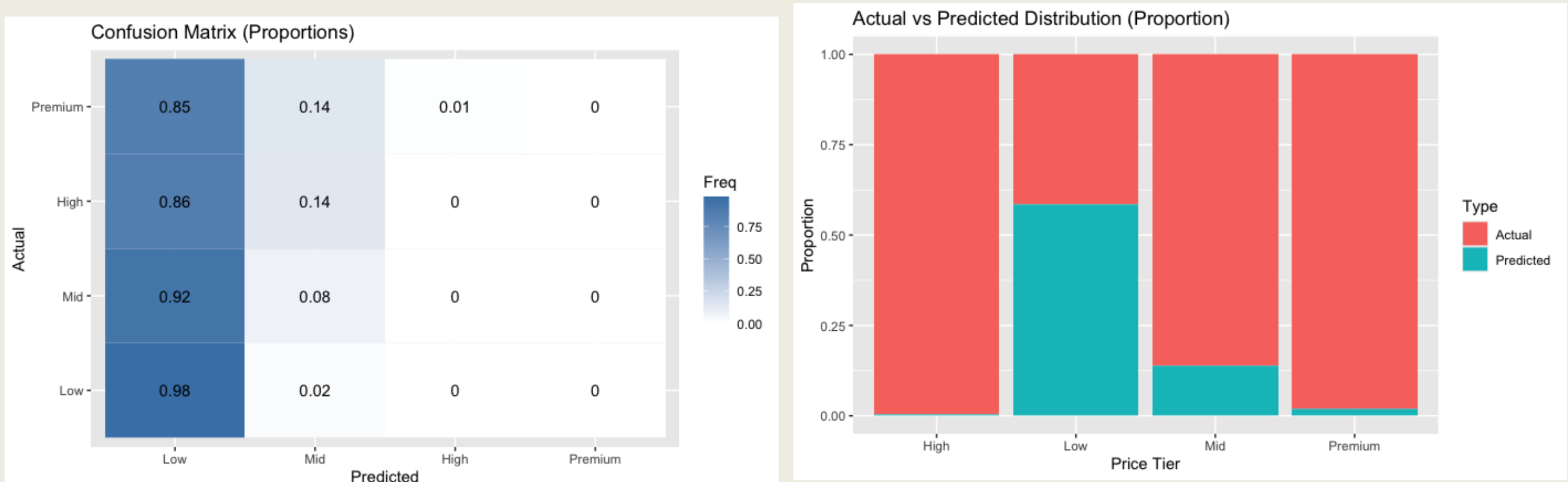
Games binned into 4 price tiers (Low / Mid / High / Premium). Fit via *MCMCmnl* (MCMCpack): 5,000 iterations, 1,000 burn-in, thinning = 5. Frequentist multinomial (nnet) used for test-set predictions.

Results (Model 2)

Overall accuracy: 68.3% barely above naive majority-class baseline (67.6% Low tier). Model largely predicted Low for everything.

- Low: 98% correct · Mid: 92% correct
- High: 0.2% correct · Premium: 0% correct

Posterior findings: positive reviews → cheaper games ($\beta \approx -0.38$ for Premium);
peak CCU → more likely Premium; language count → small positive effect on all tiers.



Conclusion

- A single MLP cannot span the full Steam price range; segmenting by price regime is key (R² 0.564).
- The Bayesian model's accuracy was dominated by the Low tier, but its posteriors revealed that engagement breadth (peak CCU, language count) predicts premium pricing more than review volume alone.
- Both models suggest price tier classification is a more natural framing than regression for this problem.

Next Steps

- Frame as classification from the start
- Add publisher-level embeddings for brand effects
- Replace full-batch GD with mini-batch / PyTorch for deeper nets